H

millennia institute

Class: Adm No

Candidate Name: _____

2026 Preliminary Examinations
Pre-University 3

GEOGRAPHY (HIGHER 2) **9173/02**
Paper 2 **14 September 2026**

3 hours

Additional Materials: Insert, Answer Booklet(s)

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front of the answer booklet. If you need additional writing paper, ask the invigilator for a continuation booklet.

Section A
Answer **both** questions.

Section B
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 4 Fieldwork

1	A group of 15 students decided to investigate the local community's response to climate change in Rotterdam, the Netherlands. Rotterdam is a low-lying city that is increasingly vulnerable to intense rainfall and pluvial flooding. Various measures, including water squares, green spaces, permeable surfaces and stormwater-storage facilities, have been introduced to improve the city's resilience. The students selected five locations in Rotterdam: Water Square Benthemplein, Rotterdam Centraal and Kruisplein, Noordplein, Dakpark Rotterdam and Museumpark. The students agreed to investigate the following hypothesis: ***The extent of public climate-adaptation measures implemented by the Municipality of Rotterdam is greater than the extent of household climate-adaptation measures adopted by individuals in Rotterdam.*** The students conducted secondary research online for basic information about the five locations and the climate-adaptation measures found at each location. They designed a questionnaire to investigate people's awareness of, support for and adoption of climate-adaptation measures. They also produced an environmental data-collection sheet to record the climate-adaptation features observed at each location. The students used convenience sampling to survey 25 people at each location. They conducted their fieldwork from 11 am to 1 pm on two weekdays in October. It was cool and rainy on both days. At each location, the students carried out the following steps: • They approached people who appeared to be at least 18 years old. • They verified whether each person lived, worked or studied in Rotterdam. • They asked willing participants to complete the questionnaire. • They identified climate-adaptation features within 50 metres of the survey point. • They used the environmental data-collection sheet to give each feature a score based on its perceived effectiveness in managing intense rainfall. In total, the students surveyed 125 people. Resource 1 shows the locations of the five fieldwork sites in Rotterdam. Resource 2 shows basic information about the five locations and their climate-adaptation measures. Resource 3 is a photograph of Benthemplein Water Square during dry and wet weather. Resource 4A shows the survey questionnaire used by the students. Resource 4B shows the environmental data-collection sheet used by the students. Resource 5 shows the results collected for environmental perception survey during the investigation.

(a)	Explain the suitability of the students' hypothesis for investigating the Rotterdam community's response to climate change.	[4]
	Award 1 mark for each valid explained point, up to a maximum of 4 marks. *For full 4 marks, responses need to include all 3 criteria.* • *how clearly the independent and dependent variables are defined;* • *whether the investigation is manageable;* • *whether the variables are measurable or researchable.* **Well-defined variables** • The independent variable is the **type of climate response**, namely public climate-adaptation measures or household climate-adaptation measures. **[1]** • The dependent variable is the **extent of implementation or adoption**, measured through support for public measures and adoption of household measures. **[1]** **Manageability** • The investigation is manageable because 15 students can divide the five sites among themselves and survey 25 respondents at each site. **[1]** • The spatial scale may be less manageable because the five sites are spread across Rotterdam, requiring additional travelling time and coordination between the students. **[1]** • Surveying 125 respondents within four hours over two weekdays may be difficult, particularly if some people refuse to participate. **[1]** **Measurability and researchability** • Support for public adaptation can be measured quantitatively using the five-point Likert-scale questions in the questionnaire. **[1]** • Household adoption can be measured by counting the number of adaptation measures reported by each respondent. **[1]** • Qualitative explanations can be collected through the open-ended question, helping the students understand reasons for respondents' actions. **[1]**	AO2
(b)	Explain the limitations of the sampling method used by the students.	[4]
	• *Award 1 mark for identifying a valid limitation.* • *Award an additional mark for a contextualised explanation.* • Convenience sampling is likely to produce a biased sample. [1] ○ The students may approach people who appear friendly or are easy to reach, so not every member of the Rotterdam community has an equal chance of selection. [+1] • Surveying from 11 am to 1 pm on weekdays may produce an unrepresentative sample. [1] ○ Full-time workers may be underrepresented, while students, retirees and people not working at that time may be overrepresented. [+1] • The cool and rainy weather may affect who is available for selection. [1] ○ Fewer people may remain outdoors, and those who do may be unwilling to stop, reducing the size and diversity of the potential sample. [+1] • Surveying only people present at the five selected sites may create location bias. [1] ○ Users of climate-adaptive public spaces may be more aware of or supportive of climate adaptation than other members of the Rotterdam community. [+1] • The students surveyed the same number of people at every site, small sample size per site. [1] ○ The five locations have different numbers and types of users, so 25 respondents at each site may not represent their relative populations. [+1]	AO1

(c)	With reference to Resources 1 and 3, explain how the students could minimise potential risks when undertaking the fieldwork.	[6]
	• Award 1 mark for identifying and contextualising a valid risk • Award additional mark for an appropriate mitigation measure.	AO1

Risk [1]	**Mitigation [+1]**
Wet steps, sloping surfaces and the edges of the sunken basins at Benthemplein may be slippery, causing students to fall	Students should wear footwear with good grip, remain on designated paths and avoid entering the basins during wet weather.
The basins at Benthemplein may fill with water during intense rainfall, potentially trapping or endangering students inside them.	Students should check the weather forecast, leave the basins immediately when rainfall intensifies and conduct observations from a safe raised position.
Students may encounter traffic, bicycles or fast-moving pedestrians when travelling between the five locations.	They should use designated pedestrian crossings and paths, remain alert and avoid obstructing busy routes while conducting questionnaires.
Cool and rainy weather may cause students to become wet and cold.	Students should wear waterproof clothing, bring umbrellas and take breaks in sheltered places.
Some members of the public may respond aggressively or make students feel unsafe during the questionnaire survey.	Students should work in pairs or groups, conduct surveys in visible public areas and withdraw immediately if a person becomes hostile.
The group may become separated while travelling between fieldwork locations.	Students should remain in assigned groups, carry charged mobile phones and agree on meeting points and emergency contact procedures.

(d)	Using Resources 4A and 4B, explain the strengths and limitations of the survey questionnaire and environmental perception survey.	[6]
	• *Award 1 mark for identifying a valid point* • *Award an additional mark for a contextualised explanation.* • *Responses should cover both strengths and limitations and refer to both Resources 4A and 4B.*	AO2

Strengths

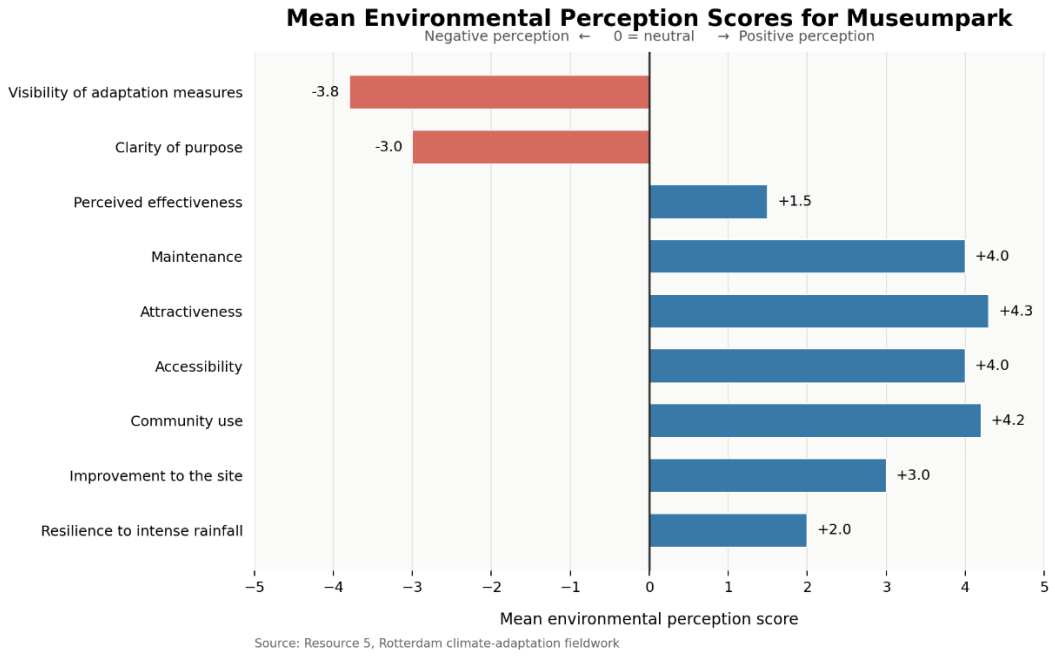
- The questionnaire mainly uses closed questions. [1]
  - This allows responses from 125 people to be collected quickly, coded numerically and compared across the five locations. [+1]
- Questions 6 and 8 are relevant to the hypothesis. [1]
  - They measure support for public climate-adaptation measures and adoption of household measures respectively. [+1]
- Question 10 is open-ended to collects qualitative data [1]
  - This does not restrict respondents to predetermined options, which allows respondents to suggest community responses not anticipated by the students. [+1]
- The environmental perception survey uses the same −5 to +5 scale at every location. [1]
  - This produces quantitative data that can be totalled and compared consistently across the five sites. [+1]

	Limitations • Question 4 uses jargon such as “pluvial flooding” and “blue-green infrastructure”. [1] ○ Respondents may misunderstand the terms, reducing the validity of their answers. [+1] • Question 7 is a leading question, which may lead to biased results. [1] ○ Stating that the measures reduce flooding and improve residents’ lives may encourage respondents to answer “yes”. [+1] • The environmental perception criteria are subjective. [1] ○ Students may interpret terms such as “effective”, “attractive” and “resilient” differently, reducing consistency between observers. [+1] • Underground adaptation measures cannot be observed easily. [1] ○ Kruisplein and Museumpark may receive low perception scores even though their underground storage systems may effectively reduce flooding. [+1] • The environmental survey records the students’ perceptions. [1] ○ It does not directly measure the local community’s perceptions and therefore may not fully address the investigation’s hypothesis. [+1] • The questionnaire is available only in English. [1] ○ Respondents who are not proficient in English may misunderstand the questions or be unable to participate, reducing the accuracy and representativeness of the data collected. [+1] • The questionnaire lacks an introduction explaining its purpose and how the data will be handled. [1] ○ Without assurances of anonymity and confidentiality, respondents may be reluctant to participate or answer sensitive questions about income and household spending honestly, reducing the reliability and validity of the results. [+1]	
(e)	Explain how students could improve the accuracy and reliability of their data collection procedures during the geographical investigation.	[5]
	• *Award 1 mark for identifying a suitable procedural improvement* • *Award an additional mark to explain how it improves accuracy **AND** reliability.* **Accuracy** • Students could pilot the questionnaire before conducting the main survey. [1] ○ This would identify unclear, technical or leading questions, such as “pluvial flooding” and “blue-green infrastructure”. These questions could then be revised so that respondents understand them correctly and their answers more accurately reflect their actual knowledge and opinions. [+1] • Students could check every completed questionnaire and environmental perception sheet while still at the fieldwork location. [1] ○ This would allow missing responses, incorrectly recorded values and scores outside the permitted range to be identified and corrected, making the recorded data closer to the respondents’ intended answers and the students’ actual observations. [+1] • Students could support their environmental perception scores with objective evidence, such as fixed-point photographs and counts of adaptation features. [1] ○ This would allow observations to be checked after the fieldwork and reduce inaccurate judgements about the presence, visibility or condition of climate-adaptation measures. [+1] • Convenience sampling may over-represent readily available respondents and under-represent other demographic groups. [1] ○ Using stratified, quota or systematic sampling would reduce sampling bias and demographic skew, making the findings more accurately representative of Rotterdam’s community. [+1]	A O1

Reliability

- All students could receive the same training and follow a standardised data-collection procedure. [1]
  - For example, they should use the same introduction, ask questions in the same order and survey the same 50-metre area. This would reduce procedural differences and make it more likely that another group would obtain similar results. [+1]
- Clear descriptors could be provided for every score on the −5 to +5 environmental perception scale. [1]
  - This would reduce differences in how students interpret terms such as “effective”, “attractive” and “resilient”, producing more consistent scores between different observers. [+1]
- The data collection could be repeated at each location on different days and at different times. [1]
  - Comparing repeated results would show whether the findings are consistent rather than being caused by unusual weather or pedestrian activity on one occasion. Calculating a mean from the repeated surveys would also reduce the influence of anomalous results. [+1]

(f)	Suggest how the ‘mean environmental perception scores for Museumpark’ shown in Resource 5 may be represented.	[5]
	Mark allocation: • Uses a suitable bipolar bar graph, with horizontal bars extending from zero. [1] • Provides an appropriate title and labels the axis as the mean environmental perception score. [1] • Uses an accurate, even scale from −5 to +5, with zero clearly identified at the centre. [1] • Correctly labels the nine environmental-perception elements. [1] • Accurately plots the data, with negative values extending left and positive values extending right. [1]	AO2



(g)	Using all the Resources, evaluate the validity of the data collected in determining whether the students' hypothesis about public implementation and household adoption is supported.	[10]

Indicative content:

Level/ Marks	Understanding of validity	Use of resources	Analysis and evaluation	Judgement
L3 8–10	Demonstrates clear and accurate understanding that validity concerns whether the data measure community support and household adoption as stated in the hypothesis.	Use specific and accurate evidence from several resources. References are integrated into the explanation rather than merely named.	Provides developed explanations of both strengths and limitations. Consistently links methodological features to their effect on validity. Recognises the relative importance of different issues.	Reaches a clear, supported and balanced judgement about whether the data are valid for determining if the hypothesis is supported.
L2 4–7	Demonstrates some understanding of validity, although it may occasionally be confused with accuracy, reliability or representativeness.	Use some relevant evidence from the resources, but references may be general, uneven or insufficiently developed.	Explain some strengths or limitations and make some links to validity. Evaluation may be unbalanced, repetitive or focused mainly on describing methods.	Provides a partial or general judgement, which may not be fully supported by the preceding discussion.
L1 1–3	Demonstrates limited understanding of validity. May treat evaluation as a list of generic advantages and disadvantages.	Makes little or no meaningful reference to the resources or hypothesis.	Identifies simple strengths or limitations with little explanation of how they affect validity. Response is mainly descriptive.	Gives no judgement or an unsupported assertion.

- *Validity concerns whether the data collected actually measure and allow a meaningful comparison between:*
  - *the extent of public climate-adaptation measures implemented by the Municipality of Rotterdam; and*
  - *the extent of household climate-adaptation measures adopted by individuals in Rotterdam.*
- *Markers should credit strengths and limitations only when candidates explain how they affect the validity of the data for testing the revised hypothesis.*

Suggested responses:

Strengths

Public implementation

- Information about the five sites identifies public measures that have been implemented, including water squares, green spaces and underground reservoirs. This provides direct evidence of public implementation.
- Storage capacities at Benthemplein and Museumpark provide quantitative evidence about the scale of some public measures, increasing validity.
- The five sites contain different types of adaptation measures, allowing the students to investigate a range of public responses.

Household adoption

- Question 8 directly asks respondents which household adaptation measures they have adopted. This is relevant to the household-adoption part of the hypothesis.
- Several household measures are included, such as green roofs, rain barrels, vegetation and flood insurance. This gives a broader indication of adoption.
- Question 9 records household expenditure, providing additional evidence about the extent of individual investment in adaptation.
- Surveying 125 people across five locations provides a reasonable range of household responses.

Limitations

Public implementation

- The environmental perception survey measures students' views of visibility, attractiveness and effectiveness. These do not directly measure the number or scale of measures implemented by the authorities.
- Underground measures at Kruisplein and Museumpark may receive low scores because they cannot be seen, even though they have been implemented and may store large volumes of water.
- The five sites were selected because they contain known adaptation measures. They may overrepresent public implementation and may not represent Rotterdam as a whole.
- Consistent quantitative information, such as storage capacity or area covered, is not provided for all five sites. This limits valid comparison of public implementation.

Household adoption

- Household adoption is self-reported and cannot be verified, so respondents may overstate or inaccurately recall the measures adopted.
- Question 8 has no "none of the above" option. A blank response could mean no adoption or an unanswered question.
- Adoption may depend on housing type, ownership and income. Tenants and apartment residents may be unable to install measures even if they support climate adaptation.
- Respondents who only work, study in or are visiting Rotterdam may not have households in the city, making their adoption data less relevant.
- Convenience sampling during rainy weekday lunchtimes may not represent individuals across Rotterdam.

Problems comparing the two parts

- Public implementation and household adoption are measured using different methods and units.
- A large public reservoir cannot be directly compared with one household measure such as a rain barrel.
- "Much greater" is not quantified, so the difference needed to support the hypothesis is unclear.
- Public-support questions measure attitudes towards public measures rather than the authorities' actual implementation.
- Only environmental perception results are provided. Household-adoption results are missing, so the two parts of the hypothesis cannot be compared.

Possible improvements

Improving the validity of public-implementation data

- Use official city-wide records to identify the number and distribution of adaptation measures implemented across Rotterdam. This would be more representative than using only five selected sites.
- Select sites using stratified sampling, including locations with high, moderate and low levels of adaptation. This would reduce the overrepresentation of prominent adaptation projects.
- Measure implementation using objective indicators, such as the number of measures, area covered and water-storage capacity, instead of students' perceptions of attractiveness or visibility.
- Collect the same quantitative information at every site. For example, record storage capacity and area covered at all five locations to allow valid comparison.
- Confirm that each measure was implemented by the Rotterdam authorities rather than by private organisations or community groups.

Improving the validity of household-adoption data

- Survey only people who live in Rotterdam or analyse residents separately from workers, students and visitors. This would ensure that respondents' household actions are relevant to the hypothesis.
- Use stratified sampling based on neighbourhood, age, housing type and tenure. This would produce a more representative sample of Rotterdam households.
- Add "none of the above" to Question 8. This would distinguish non-adoption from a question that was accidentally left unanswered.
- Ask whether respondents own or rent their homes and whether they live in a house or apartment. This would help distinguish unwillingness to adopt measures from an inability to make changes.
- Ask respondents why they adopted each measure. This would establish whether actions such as planting vegetation were undertaken specifically for climate adaptation.
- Where practical, corroborate self-reported adoption using photographs, receipts or follow-up interviews, with respondents' consent.

Improving the comparison

- Convert public implementation and household adoption into comparable standardised indices, such as scores from 0 to 10.
- Define "much greater" by stating the minimum difference required for the hypothesis to be supported.
- Provide the actual results for household adoption and compare them with the public-implementation results.
- Compare similar functions where possible, such as the amount of stormwater stored by public facilities with the amount stored by household measures.
- Use both quantitative data and interview evidence to explain why differences exist between public implementation and household adoption.

Suggested Level 3 response

Overall, the data have limited validity for determining whether the hypothesis is supported. Some resources provide relevant evidence of public implementation and the questionnaire could measure household adoption. However, the two variables are measured differently, and the results needed to compare them are incomplete.

For public implementation, Resource 2 identifies measures such as the water-storage basins at Benthemplein and underground reservoirs at Kruisplein and Museumpark. The storage capacities of approximately 1.7 million litres at Benthemplein and 10 million litres at Museumpark provide quantitative evidence that substantial public measures have been implemented. Resource 3 also shows Benthemplein storing water during wet weather. However, the five sites in Resource 1 appear to have been selected because they contain known adaptation measures. They may therefore overrepresent public implementation and may not represent Rotterdam as a whole. In addition, Resource 4B measures students' perceptions of visibility, attractiveness and community use rather than the actual extent of implementation. For example, Museumpark received a visibility score of −3.8 in Resource 5 despite having a large underground reservoir. This reduces validity because measures that cannot be seen may be underestimated.

For household adoption, Question 8 in Resource 4A directly asks respondents which measures they have adopted, while Question 9 records their expenditure. These questions are relevant to the household part of the hypothesis. However, adoption is self-reported and cannot be verified. Question 8 also lacks a "none of the above" option. Furthermore, tenants and apartment residents may be unable to install green roofs or permeable paving. As housing tenure is not recorded, low adoption may reflect an inability to act rather than weak individual response. Convenience sampling during rainy weekday lunchtimes may also underrepresent full-time workers, while respondents at Rotterdam Centraal may not live in the city.

The main limitation is that public implementation and household adoption are not directly comparable. A large public reservoir cannot be treated as equivalent to one household rain barrel, and "much greater" is not quantified. Most importantly, Resource 5 provides only environmental perception results and not the household-adoption results from Questions 8 and 9. The students should use official city-wide implementation data, survey only Rotterdam residents, record housing tenure and convert both variables

	into comparable indices. Therefore, although the resources show that public measures exist and the questionnaire could collect household data, the available evidence is insufficient to support the hypothesis validly.	

Cluster 3 Sustainable Future and Climate Change

2	Resource 6 shows the average carbon dioxide emissions per capita across the world in 2019. Resource 7 shows the share of plastic waste management methods of Europe and Asia in 2019. Resource 8 shows a graph of atmospheric CO_2 (parts per million, NOAA) and Global Temperature Anomaly (°C, GISS) from 1964 to 2008. Resource 9 is a photograph of an aerial view of the Bengal Delta in Bangladesh in April 1991.		
	(a)	Using Resource 6, suggest reasons for the variations in the methods of plastic waste management between Europe and Asia in 2019.	[6]
		• *Award 1 mark for suggesting a plausible reason.* • *Award an additional mark for using relevant comparative data from Resource 6 and explaining how the identified reason could account for the observed differences.* • Europe may have greater financial capacity to invest in more costly plastic waste-treatment infrastructure, such as incineration and waste-to-energy plants. [1] ○ This could explain why 38% of plastic waste was incinerated in Europe compared with only 19% in Asia, as greater government and private-sector funding enables the construction and operation of technologically advanced incineration facilities, increasing the proportion of waste that can be treated this way. [+1] • Europe generally has a higher level of development, with more extensive formal waste-collection, sorting and treatment infrastructure than less-developed parts of Asia. [1] ○ This could explain why only 6% of plastic waste was mismanaged in Europe compared with 34% in Asia, as more developed and accessible waste-management systems enable plastic waste to be formally collected and treated, reducing uncontrolled dumping, open burning or disposal into waterways. [+1] • Differences in political governance may affect the effectiveness of waste-management policies and their enforcement. [1] ○ Stronger regulatory capacity and enforcement in many European countries may encourage proper disposal and recycling, helping to explain why 12% of plastic waste was recycled in Europe compared with 8% in Asia, while mismanaged waste was much lower at 6% compared with 34%; effective enforcement can reduce illegal disposal and channel more waste into formal treatment systems. [+1]	
	(b)	Using Resource 7, explain the impact of multiple deprivations experienced by slum dwellers on sustainable urban development.	[6]
		• *Award 1 mark for explaining an impact of deprivation experienced by slum dwellers.* • *Award an additional mark for explaining how the identified impact affects social, economic and/or environmental sustainability.* • Slum dwellers may experience inadequate access to proper waste collection and sanitation services, resulting in large amounts of solid waste being dumped openly around their homes, as shown in Resource 7. [1]	

		○ This undermines environmental sustainability as accumulated waste can contaminate soil and water and block drains, while poor sanitation may also increase exposure to disease, reducing the ability of the city to provide a clean and healthy living environment over time. [+1] • Slum dwellers may experience poor-quality and insecure housing, as Resource 7 shows makeshift and dilapidated structures constructed from low-quality materials. [1] ○ This undermines social sustainability because residents may be more vulnerable to hazards such as fires, storms or structural collapse, reducing their safety and quality of life and contributing to persistent inequalities in access to adequate housing. [+1] • Living close to uncollected rubbish exposes slum dwellers to disease vectors such as rats, flies and mosquitoes. [1] ○ A greater incidence of disease can reduce residents' health and quality of life and widen inequalities in living conditions, undermining the goal of providing a safe, healthy and equitable urban environment. [+1] • Slum dwellers may experience poverty and limited access to basic infrastructure and services, evident from the poorly developed surroundings and lack of formal infrastructure in Resource 7. [1] ○ This undermines economic sustainability because poor health and living conditions may reduce residents' ability to work productively and increase public expenditure needed to improve healthcare, housing and infrastructure, limiting resources available for the city's long-term development. [+1] • Poor access to education may limit slum dwellers' acquisition of skills and qualifications. [1] ○ This may restrict access to stable and higher-paying formal employment, perpetuating poverty and informal employment and limiting the contribution of residents to inclusive and sustained urban economic growth. [+1]	
	(c)	Citing data from Resource 8, describe the relationship between changes in atmospheric CO_2 concentration and global temperature anomaly over time.	[4]
		• *Award 1 mark for describing a valid relationship.* • *Award an additional 1 mark for supporting the identified relationship with appropriate and accurate data from Resource 8.* • There is an overall positive relationship between atmospheric CO_2 concentration and global temperature anomaly, as both generally increased between 1964 and 2008. [1] ○ Atmospheric CO_2 concentration increased from about 320 ppm in 1964 to about 385 ppm in 2008, while global temperature anomaly generally increased from about −0.2°C to +0.4°C over the same period. [+1] • However, the relationship is not perfectly consistent in the short term, as temperature anomaly fluctuated considerably even though atmospheric CO_2 concentration increased relatively steadily. [1]	

		○ For example, from about 1988 to 1992, CO_2 concentration continued to rise from about 350 ppm to 356 ppm, while temperature anomaly fell from about +0.3°C to +0.1°C. [+1] • Despite the peaks and troughs in global temperature anomaly line graph, there is little to no direct influence on the steady gradient of the atmospheric CO_2 concentration. [1]	
	(d)	Explain reasons for uncertainty over the present and future impacts of contemporary climate change.	[4]
		• *Award 1 mark for identifying a valid reason for uncertainty.* • *Award an additional mark for elaborating how the identified reason creates uncertainty over the magnitude, timing and/or spatial extent of climate-change impacts, or using an appropriate example to explain the uncertainty.* • [Lack of full understanding of physical processes] The climate system involves complex interactions and feedback mechanisms between the atmosphere, oceans, cryosphere and biosphere that are not fully understood. [1] ○ For example, uncertainty over the strength of ice-albedo, cloud and carbon-cycle feedbacks means climate models may produce different estimates of the magnitude, timing and spatial distribution of warming and its associated impacts. [+1] • [Incomplete in-situ data] There is limited direct climate data from vast or inaccessible areas, particularly oceans, deserts and polar regions, because monitoring stations are sparse and difficult or costly to maintain. [1] ○ These data gaps make it harder to establish accurate baseline conditions and long-term trends, creating uncertainty over the magnitude and spatial extent of present changes and reducing confidence in projections of future impacts. [+1] • [Intrinsic measurement errors in current climate data] Climate measurements have a margin of error due to limitations in measuring instruments and data collection methods. [1] ○ Such uncertainties in measurements of temperature, precipitation or sea level can affect estimates of the rate and magnitude of observed change, making it more difficult to determine precisely the extent of present impacts and to calibrate projections of future impacts. [+1] • [Uncertainty over future greenhouse gas emissions] Future greenhouse gas emissions depend on unpredictable human factors, such as population and economic growth, energy consumption, technological change and the effectiveness of climate policies. [1] ○ Different future emission pathways would result in different degrees of warming; for example, rapid decarbonisation could limit warming whereas continued high fossil-fuel use could increase it, creating uncertainty over the severity of future impacts such as sea-level rise, heatwaves and changing precipitation patterns. [+1]	
	(e)	Using Resource 9, explain the impacts of coastal flooding on different groups of people and the environment in Bangladesh.	[4]

		• *Award 1 mark for explaining a valid impact of coastal flooding on a specific group of people or the environment, with reference to Resource 9 where appropriate. [1]* • *Award an additional mark for developing the impact by explaining the consequences and/or differential vulnerability of the affected group or the wider effects on the environment.* • Poorer households living in low-lying coastal areas may experience damage to or loss of their homes and possessions when coastal flooding occurs, as Resource 9 shows settlements surrounded by extensive floodwaters. [1] ○ Poorer households may be disproportionately affected because they are more likely to live in less resilient housing and have limited savings, insurance or resources to repair or rebuild their homes, potentially causing prolonged displacement and making recovery from flooding more difficult. [+1] • Farmers may experience crop losses as large areas of agricultural land are submerged by floodwater, as shown in Resource 9. Saltwater intrusion may also increase soil salinity and reduce agricultural productivity even after the floodwaters recede. [1] ○ Poorer farmers who depend heavily on agriculture and have limited savings or alternative livelihoods may be more vulnerable, as crop losses reduce both their food supply and income, while they may lack the financial resources to recover compared with households with diversified or higher incomes. [+1] • Fishermen may experience disruption to their livelihoods if flooding damages boats, fishing equipment or access to coastal fishing areas. [1] ○ Lower catches or the loss of fishing assets may reduce household income, with poorer fishers particularly vulnerable because they may have limited savings or alternative sources of employment to support recovery. [+1] • Coastal flooding may erode soils and damage coastal vegetation and habitats. [1] ○ The resulting habitat degradation may reduce biodiversity and ecosystem functioning, while the loss of coastal vegetation can reduce natural protection against subsequent flooding and erosion. [+1] • Coastal flooding can cause saltwater intrusion into soils and freshwater sources, damaging vegetation and reducing the availability of usable agricultural land and freshwater. [1] ○ Floodwater may also carry sewage and other pollutants into surrounding water bodies, degrading water quality and aquatic ecosystems. This can further affect communities dependent on these environmental resources for farming, fishing and water supply. [+1]	
	(f)	With the use of examples, explain the impacts of changes in aquatic ecosystems brought about by climate change on humans.	[6]
		• *Award 1 mark for explaining a change in an aquatic ecosystem brought about by climate change (change in temp/ppt patterns), supported by an appropriate example where relevant.* • *Award an additional mark for explaining how the identified change in the aquatic ecosystem affects humans.*	

		• Rising sea temperatures can cause coral bleaching and mortality, reducing the health and biodiversity of coral reef ecosystems. [1] ○ For example, repeated mass bleaching events on Australia's Great Barrier Reef can reduce the abundance and diversity of reef fish. This may lower catches and income for communities dependent on reef fisheries, while degradation of the reef can also reduce tourism revenue and employment associated with diving and snorkelling. [+1] • Ocean warming can cause commercially important fish species to shift their geographical distribution towards cooler waters or greater depths. [1] ○ For example, warming waters in the North Atlantic have contributed to shifts in the distribution of Atlantic mackerel, changing where fish can be caught. Fishers in areas where stocks decline may face lower catches, reduced income and greater costs from travelling farther to fishing grounds, threatening fishing-dependent livelihoods. [+1] • Rising sea temperatures and sea levels can damage coastal aquatic ecosystems such as mangroves, especially where mangroves cannot migrate inland because of human development. [1] ○ For example, mangrove degradation in low-lying coastal areas can reduce their ability to dissipate wave energy and provide natural coastal protection, increasing the exposure of coastal communities and infrastructure to erosion, storm surges and flooding and raising the costs of coastal protection and recovery. [+1] • Increasing atmospheric CO_2 causes ocean acidification, which reduces the ability of shell-forming organisms such as oysters and mussels to build and maintain calcium carbonate shells. [1] ○ For example, oyster hatcheries along the Pacific Northwest coast of the USA have experienced difficulties associated with corrosive seawater conditions. Reduced shellfish survival can decrease production and threaten the incomes and employment of people dependent on aquaculture and shellfish industries. [+1] • Warming water can alter aquatic food webs and reduce suitable habitats for some fish species. [1] ○ In regions where communities depend strongly on fisheries for food, declining fish populations may reduce the availability of fish as a source of protein and increase food insecurity, particularly among lower-income households with fewer alternative food sources. [+1]	

Section B

Answer **either** Question 3 **or** Question 4.

	Cluster 3 Sustainable Future and Climate Change	
3	'Liveability is more complementary than it is in tension with sustainable urban development.' Evaluate this statement.	[20]
	Indicative Content Candidates are required to demonstrate contexts that show complementarity between liveability and sustainable urban development, but may also generate tensions because urban resources and the benefits of development are unevenly distributed. Responses could consider how the needs of vulnerable groups, particularly the elderly and women, are incorporated into urban planning and how this affects social well-being, accessibility, safety, participation and equity to build an inclusive society in alignment with SUD outcomes. Relevant content may include age-friendly transport and public-space design, ageing-in-place, gender-sensitive planning, mobility and care infrastructure, and the ways these contribute to both liveability and the social dimension of sustainable urban development. Students should support assertions with relevant, accurate case-study evidence and explain the geographical processes linking policies to outcomes. Higher-progress responses should move beyond describing inclusive policies to analyse the extent of complementarity and evaluate tensions arising from competing land uses, financial resources, accessibility needs and the differing interests of social groups. Stronger responses may also recognise that the relationship varies according to place, time, purpose and whose liveability is being assessed, leading to a clear and substantiated judgement about the extent to which inclusion creates complementary rather than competing outcomes.	
4	"Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming." Evaluate this statement.	[20]
	Indicative Content: *Students should evaluate whether human factors are the main cause of contemporary global warming. Responses should explain how human activities increase greenhouse-gas concentrations and strengthen the enhanced greenhouse effect, while considering how natural factors influence global temperatures. Students should compare their relative importance using criteria such as magnitude, duration, spatial scale and ability to explain the speed and persistence of contemporary warming, before reaching a reasoned judgement.* *Higher-level responses will recognise that natural factors may initiate long-term climatic changes, produce short-term variability, redistribute heat or amplify existing warming, but generally cannot explain the sustained contemporary global trend. They should distinguish between underlying forcings, feedbacks and modifiers, and recognise that human and natural factors interact. The strongest responses will conclude that human factors are dominant at contemporary global scales, while natural factors remain more significant over geological timescales or in shaping regional and short-term variations.*	

End of Paper

Copyright Acknowledgements

Resource 1	Google maps (author's own)
Resource 2	https://landezine.com/water-square-benthemplein-by-de-urbanisten/ (Last accessed 26 July 2026)
Resource 3	https://www.urbanisten.nl/work/benthemplein https://www.iurc.eu/wp-content/uploads/2022/05/Rotterdam-Presentation_compressed.pdf https://www.rotterdam.info/en/visit/finder-locations/rotterdamse-oogstmarkt https://www.rotterdam.info/en/visit/finder-locations/dakpark https://paulderuiter.nl/en/projects/parkeergarage-museumpark (Last accessed 1 August 2026)
Resource 4A and 4B	Author's own
Resource 5	Author's own
Resource 6	https://ourworldindata.org/grapher/share-plastic-fate?time=2019 (Last accessed 19 August 2026)
Resource 7	https://www.justonewayticket.com/smokey-mountain-a-walk-through-the-slums-of-manila-philippines/ (Last accessed 19 August 2026)
Resource 8	https://skepticalscience.com/the-correlation-between-co2-and-temperature.html (Last accessed 19 August 2026)
Resource 9	https://www.lse.ac.uk/granthaminstitute/publication/tackling-flooding-in-bangladesh-in-a-changing-climate/ (Last accessed 19 August 2026)